PIQA: Reasoning about Physical Commonsense in Natural Language

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arXiv:1911.11641v1 [cs.CL] 26 Nov 2019

Abstract

To apply eyeshadow without a brush, should I use a *cotton swab or a toothpick*? Questions requiring this kind of **physical commonsense** pose a challenge to today's natural language understanding systems. While recent pretrained models (such as BERT) have made progress on question answering over more *abstract* domains – such as news articles and encyclopedia entries, where text is plentiful – in more *physical* domains, text is inherently limited due to reporting bias. Can AI systems learn to reliably answer physical commonsense questions without experiencing the physical world?

In this paper, we introduce the task of physical commonsense reasoning and a corresponding benchmark dataset **Physical Interaction: Question Answering** or **PIQA**. Though humans find the dataset easy (95% accuracy), large pretrained models struggle (∼77%). We provide analysis about the dimensions of knowledge that existing models lack, which offers significant opportunities for future research.

Introduction

Before children learn language, they already start forming categories and concepts based on the physical properties of objects around them (Hespos and Spelke 2004). This model of the world grows richer as they learn to speak, but already captures *physical commonsense* knowledge about everyday objects: their physical properties, affordances, and how they can be manipulated. This knowledge is critical for day-to-day human life, including tasks such as problem solving (what can I use as a pillow when camping?) and expressing needs and desires (bring me a hard*er* pillow). Likewise, we hypothesize that modeling physical commonsense knowledge is a major challenge on the road to true AI-completeness, including robots that interact with the world and understand natural language.

Much of physical commonsense can be expressed in language, as the versatility of everyday objects and common concepts eludes other label schemes. However, due to issues of reporting bias, these commonsense properties - facts like 'it is a bad idea to apply eyeshadow with a toothpick' are rarely directly reported. Although much recent progress

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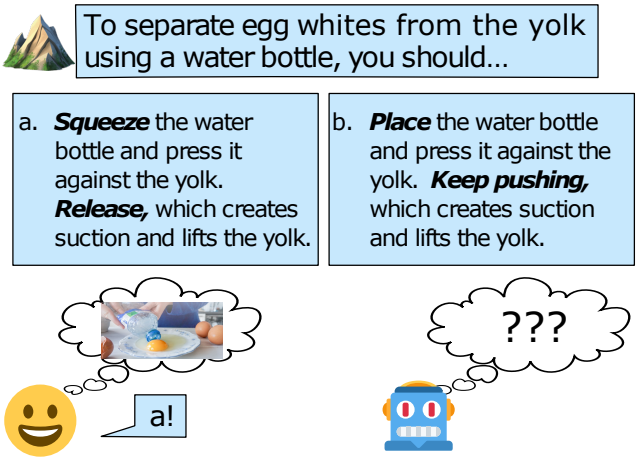

Figure 1: **PIQA**: Given a physical **goal** expressed in natural language, like 'to separate egg whites...,' a model must choose the most sensible **solution**. Our dataset tests the ability of natural language understanding models to link text to a robust intuitive-physics model of the world. Here, humans easily pick answer **a)** because separating the egg requires *pulling* the yolk out, while machines are easily fooled.

has been made in Natural Language Processing through a shift towards large-scale pretrained representations from unlabeled text (Radford et al. 2018; Devlin et al. 2019; Liu et al. 2019), the bulk of the success of this paradigm has been on core *abstract* tasks and domains. State-of-the-art models can reliably answer questions given an encyclopedia article (Rajpurkar et al. 2016) or recognize named entities (Tjong Kim Sang and De Meulder 2003), but it is not clear whether they can robustly answer questions that require physical commonsense knowledge.

To study this question and begin bridging the representational gap, we introduce **Physical Interaction: Question Answering**, or **PIQA** to evaluate language representations on their knowledge of physical commonsense. We focus on everyday situations with a preference for atypical solutions. Our dataset is inspired by instructables.com, which provides users with instructions on how to build, craft,

a. Shape, Material, and Purpose

[Goal] Make an outdoor pillow
[Sol1] Blow into a tin can and tie with rubber band ✘
[Sol2] Blow into a trash bag and tie with rubber band ✔

[Goal] To make a hard shelled taco,
[Sol1] put seasoned beef, cheese, and lettuce onto the hard shell. ✘
[Sol2] put seasoned beef, cheese, and lettuce into the hard shell. ✔

[Goal] How do I find something I lost on the carpet?
[Sol1] Put a solid seal on the end of your vacuum and turn it on. ✘
[Sol2] Put a hair net on the end of your vacuum and turn it on. ✔

b. Commonsense Convenience

[Goal] How to make sure all the clocks in the house are set accurately?

[Sol1] Get a solar clock for a reference and place it just outside a window that gets lots of sun. Use a system of call and response once a month, having one person stationed at the solar clock who yells out the correct time and have another person move to each of the indoor clocks to check if they are showing the right time. Adjust as necessary. ✘
[Sol2] Replace all wind-ups with digital clocks. That way, you set them once, and that's it. Check the batteries once a year or if you notice anything looks a little off. ✔

Figure 2: **PIQA** covers a broad array of phenomena. Above are two categories of example QA pairs. **Left** are examples that require knowledge of basic properties of the objects (flexibility, curvature, and being porous), while on the **Right** both answers may be technically correct but one is more convenient and preferable.

bake, or manipulate objects using everyday materials. We asked annotators to provide semantic perturbations or alternative approaches which are otherwise syntactically and topically similar to ensure physical knowledge is targeted. The dataset is further cleaned of basic artifacts using the `AFLite` algorithm introduced in (Sakaguchi et al. 2020; Sap et al. 2019) which is an improvement on adversarial filtering (Zellers et al. 2018; Zellers et al. 2019b).

Throughout this work we first detail the construction of our new benchmark for physical commonsense. Second, we show that popular approaches to large-scale language pretraining, while highly successful on many *abstract* tasks, fall short when a physical model of the world is required. Finally, our goal is to elicit further research into building language representations that capture details of the real world. To these ends, we perform error and corpora analyses to provide insights for future work.

Dataset

We introduce a new dataset, **PIQA**, for benchmarking progress in physical commonsense understanding. The underlying task is multiple choice question answering: given a question q and two possible solutions s_1, s_2, a model or a human must choose the most appropriate solution, of which exactly one is correct. We collect data with how-to instructions as a scaffold, and use state-of-the-art approaches for handling spurious biases, which we will discuss below.

Instructables as a source of physical commonsense

Our goal is to construct a resource that requires concrete physical reasoning. To achieve this, we provide a prompt to the annotators derived from instructables.com. The instructables website is a crowdsourced collection of instructions for doing everything from cooking to car repair. In most cases, users provide images or videos detailing each step and a list of tools that will be required. Most goals are simultaneously rare and unsurprising. While an annotator is unlikely to have built a UV-Flourescent steampunk lamp or made a backpack out of duct tape, it is not surprising that someone interested in home crafting would create these, nor will the tools and materials be unfamiliar to the average person. Using these examples as the seed for their annotation, helps remind annotators about the less prototypical uses of everyday objects. Second, and equally important, is that instructions build on one another. This means that any QA pair inspired by an instructable is more likely to explicitly state assumptions about what preconditions need to be met to start the task and what postconditions define success.

Collecting data through goal-solution pairs

Unlike traditional QA tasks, we define our dataset in terms of Goal and Solution pairs (see Figure 2 for example Goal-Solution pairs and types of physical reasoning). The Goal in most cases can be viewed as indicating a post-condition and the solutions indicate the procedure for accomplishing this. The more detailed the goal, the easier it is for annotators to write both correct and incorrect solutions. As noted above, the second component of our annotation design is reminding people to think creatively. We initially experimented with asking annotators for (task, tool) pairs via unconstrained prompts, but found that reporting bias swamped the dataset. In particular, when thinking about how to achieve a goal, people most often are drawn to prototypical solutions and look for tools in the kitchen (e.g. forks and knives) or the garage (e.g. hammers and drills). They rarely considered the literal hundreds of other everyday objects that might be in their own homes (e.g. sidewalk chalk, shower curtains, etc).

To address this, and flatten the distribution of referenced objects (see Figure 5), we prompt the annotations with links to instructables. Specifically, annotators were asked to glance at the instructions of an instructable and pull out or have it inspire them to construct two component tasks. They would then articulate the goal (often centered on atypical materials) and how to achieve it. In addition, we asked them to provide a permutation to their own solution which makes it invalid, often subtly (Figure 3). To further assist diversity

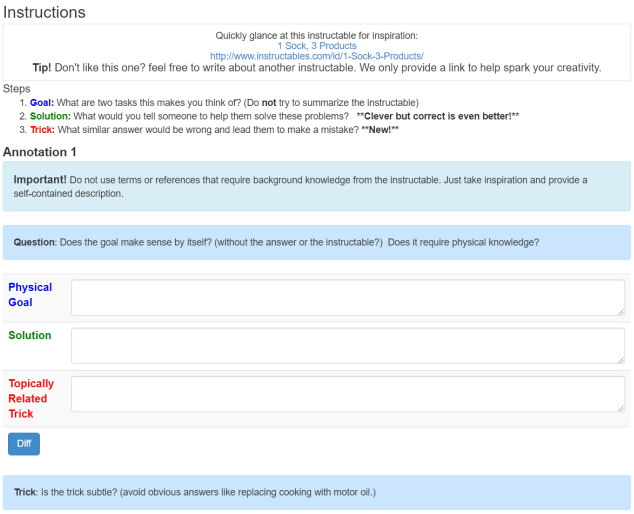

Figure 3: In the HIT design the instructable provides inspiration to think out-of-the-box (*1 Sock, 3 Products*) and annotators are asked for 1. a *physical* goal, 2. a valid solution, and 3. a trick. The trick should sound reasonable, but be wrong often due to a subtle misunderstanding of preconditions or physics. Additional HITs (not shown) were run for qualification prior to this stage and validation afterwards.[2]

we seed annotators with instructables drawn from six categories (costume, outside, craft, home, food, and workshop). We asked that two examples be drawn per instructable to encourage one of them to come later in the process and require precise articulation of pre-conditions.

During validation, examples with low agreement were removed from the data. This often meant that correct examples were removed that required expert level knowledge of a domain (e.g. special woodworking terminology) which should not fall under the umbrella of "commonsense." Because, we focus on human generated tricks, annotators were free to come up with clever ways to hide deception. Often, this meant making very subtle changes to the solution to render it incorrect. In these cases, the two solutions may differ by as little as one word. We found that annotations used both simple linguistic tricks (e.g. negation and numerical changes) and often swapped a key action or item for another that was topically similar but not helpful for completing the given goal. For this reason, our interface also includes a `diff` button which highlights where the solutions differ. This improved annotator accuracy and speed substantially. Annotator pay averaged > 15\$/hr according to both self-reporting on `turkerview.com` and our timing calculations.

[2]In addition to this design, we also include a qualification HIT which contained well constructed and underspecified (goal, solution) pairs. Annotators had to successfully (>80%) identify which were well formed to participate in the main HIT. Data was collected in batches of several thousand triples and validated by other annotators for correctness. Users will low agreement were de-qualed.

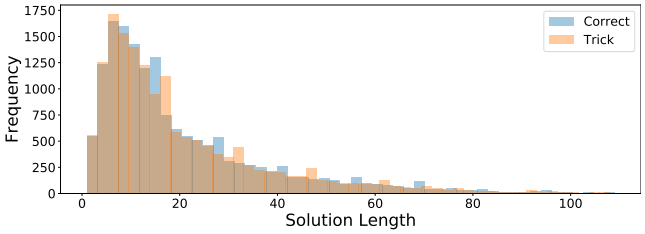

Figure 4: Sentence length distributions for both correct solutions and tricks are nearly identical across the training set.

Statistics

In total our dataset is comprised of over 16,000 training QA pairs with an additional ∼2K and ∼3k held out for development and testing, respectively. Our goals, as tokenized by Spacy,[3] average 7.8 words and both correct and incorrect solutions average 21.3 words. In total, this leads to over 3.7 million lexical tokens in the training data.

Figure 4 shows a plot of the correct and incorrect sequence lengths (as tokenized by the GPT BPE tokenizer), with the longest 1% of the data removed. While there are minor differences, the two distributions are nearly identical.

We also analyzed the overlap in the vocabulary and find that in all cases (noun, verb, adjective, and adverb) we see at least an 85% overlap between words used in correct and incorrect solutions. In total we have 6,881 unique nouns, 2,493 verbs, 2,263 adjectives, and 604 adverbs in the training data.. The most common of each are plotted in Figure 5 alongside their cumulative distributions. Again, this helps verify that the dataset revolves very heavily around physical phenomena, properties, and manipulations. For example, the top adjectives include state (*dry, clean, hot*) and shape (*small, sharp, flat*); adverbs include temporal conditions (*then, when*) and manner (*quickly, carefully, completely*). These properties often differentiate correct from incorrect answers, as shown in examples throughout the paper. We also color words according to their concreteness score (Brysbaert, Warriner, and Kuperman 2014), though many "abstract" words have concrete realizations in our dataset.

Removing Annotation Artifacts

As noted previously, we use `AFLite` (Sakaguchi et al. 2020) to remove stylistic artifacts and trivial examples from the data, which have been shown to artificially inflate model performance on previous NLI benchmarks (Poliak et al. 2018; Gururangan et al. 2018). The `AFLite` algorithm performs a systematic data bias reduction: it discards instances whose given feature representations are collectively highly indicative of the target label. In practice, we use 5,000 examples from the original dataset to fine-tune BERT-Large for this task and compute the corresponding embeddings of all remaining instances. `AFLite` uses an ensemble of linear classifiers trained on random subsets of the data to determine whether these pre-computed embeddings are strong

[3]`https://spacy.io` – all data was collected in English.